

Reservoir Simulation and Model Construction

1. Digitization of Structural and Property Contour Maps

To integrate legacy or hardcopy geological data into the Computer Modelling Group (CMG) reservoir simulator, physical contour maps must first be converted into a machine-readable digital format with the help of a specialized internal digitization software developed by **ONGC**.

The digitization process converted raw map images into **Boundary Files (.bna format)**. These files capture spatial coordinates (x, y) along with their corresponding z-values or property attributes. The following critical boundary files were successfully generated:

- **Structure Contour Map:** Defines the structural top and structural geometry of the reservoir.
- **Isopay Map:** Establishes the gross and net pay thickness distribution across the field.
- **Porosity Map:** Maps the lateral variation of matrix porosity (ϕ).
- **Initial Gas Saturation Map (S_{gi}):** Captures the spatial distribution of initial gas in place before production.

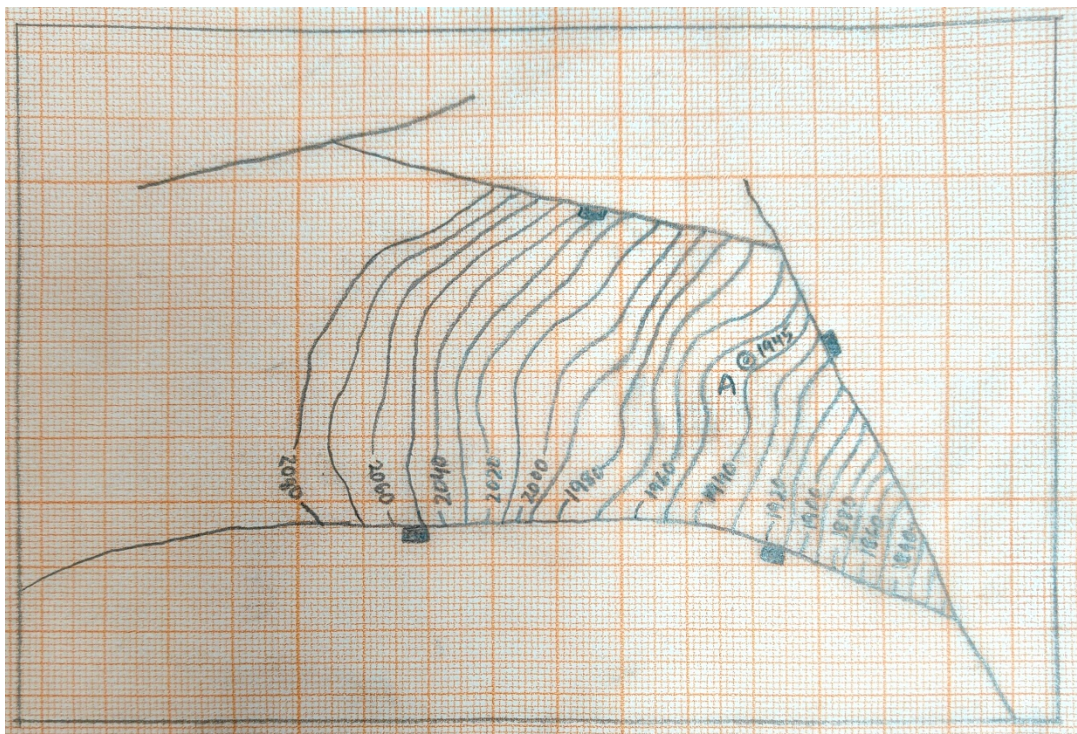


Figure 32: Structure Contour Map

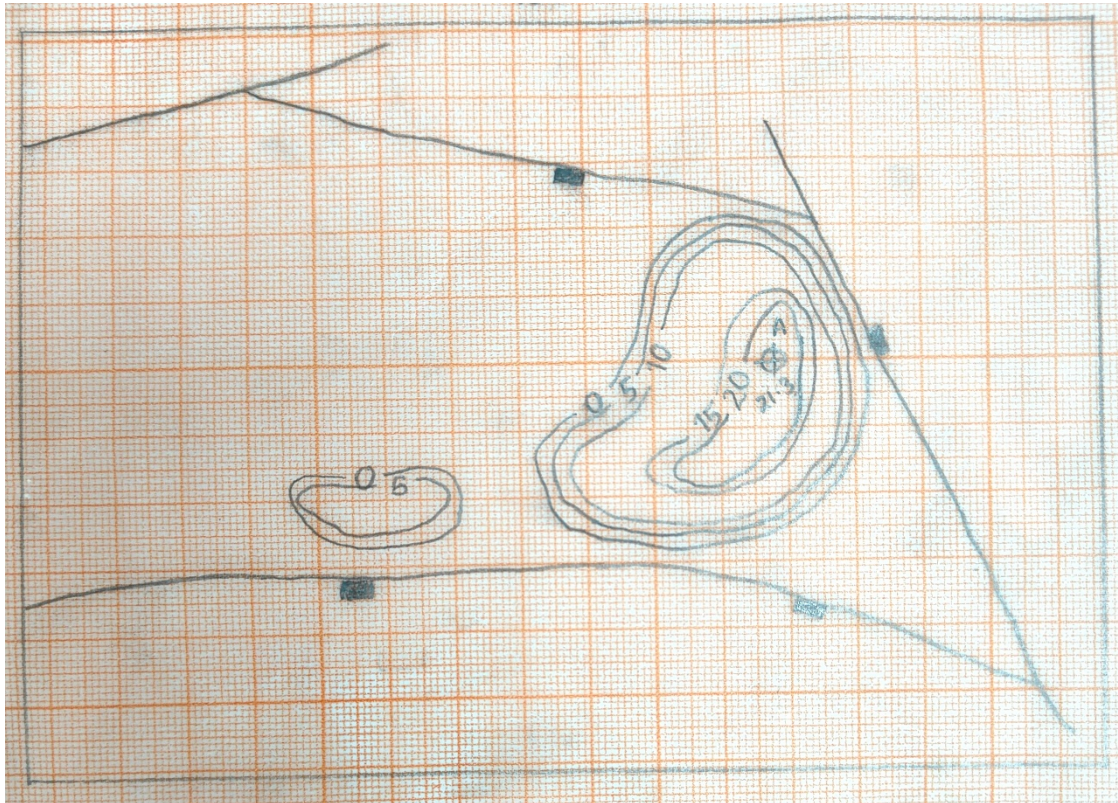


Figure 33: Isopay Map

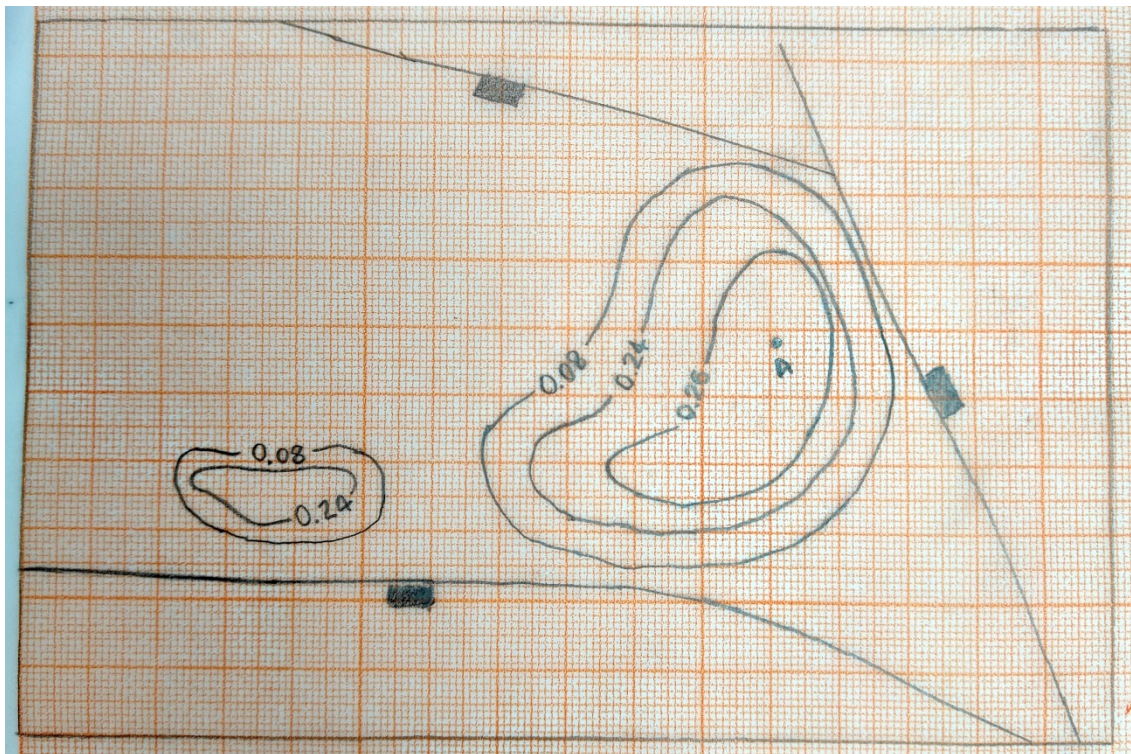


Figure 33: Porosity Map

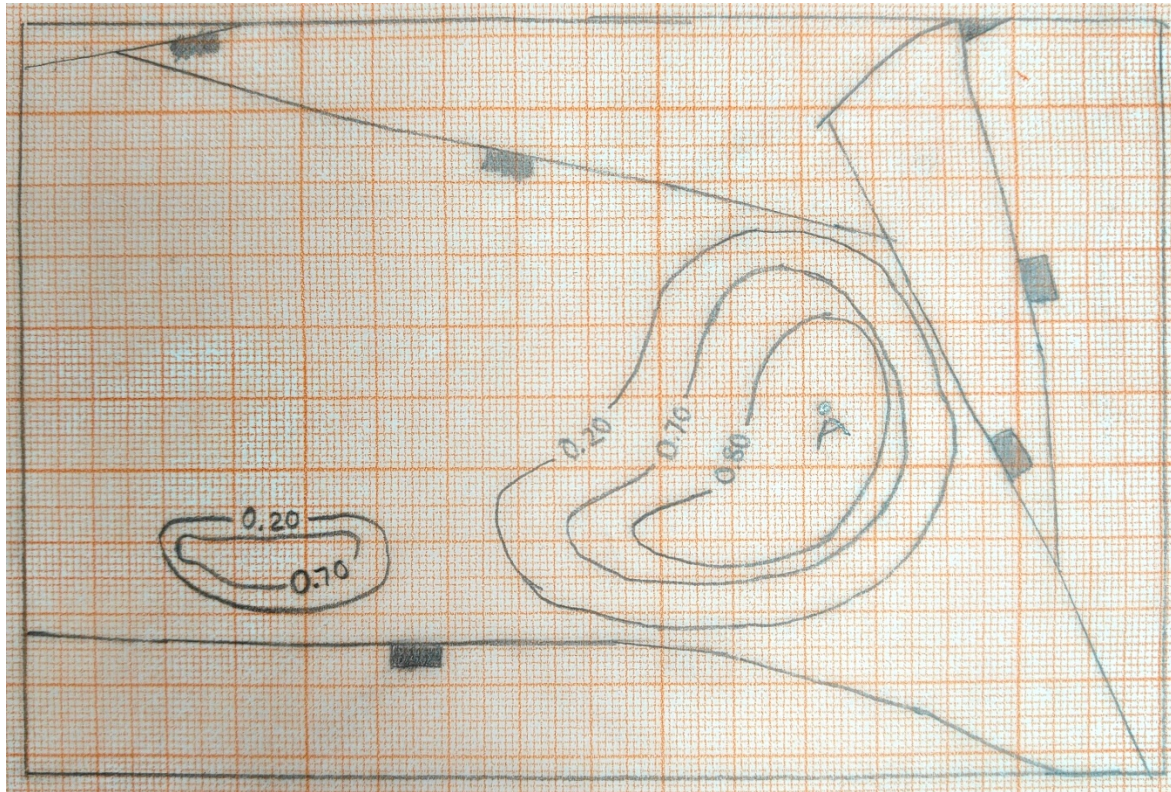


Figure 33: Initial Gas Saturation Map

2. Simulator Settings in CMG Builder

The static and dynamic model configuration was initiated within **CMG Builder**. The setup adhered to the following protocol:

- **Simulator Selection:** The **IMEX** (Black Oil Simulator) option was selected from the CMG suite as it is optimal for conventional gas reservoir behaviour under standard PVT conditions.
- **Working Units:** The **MODSI** (Modified SI) unit system was assigned.
- **Porosity Model:** A **single porosity** matrix framework was chosen.
- **Temporal Initialization:** The simulation reference start date was formally anchored at **01/04/2026**.

3. Grid Generation and Structural Alignment

- **Map Importation:** The digitized geological contour maps were imported into Builder (File → Open Map File). The file format was specified as **Atlas Boundary Format (.bna)**, with coordinate spatial units designated in meters (m).
- **Grid Framework:** A structured **Orthogonal Corner Point Grid** system was constructed via Reservoir → Create Grid.

- **Discretization Parameters:** The grid mesh was discretized into **80 blocks in the I-direction, 80 blocks in the J-direction, and 1 layer in the K-direction** (creating a 2D areal/layer model). Block dimensions were uniformly assigned as **80 * 100 m** in both the I and J spatial directions.

Model Description						
Grid Cell (X*Y*Z)			80 x 80 x 10			
Field	Fluid type	Layer	Contact	Ref. Press., KSC	Ref. Depth, m	Temp, °C
XY	GAS	10	GWC	340	1992	103

4. Spatial Property Population and Boundary Truncation

- **Property Mapping:** Using the Specify Property panel, the respective spatial maps were assigned to their corresponding simulator arrays:
 - **Grid Top:** Populated via the *Structure Contour Map*.
 - **Grid Thickness / Net Pay:** Populated via the *Isopay Map*.
 - **Porosity:** Populated via the *Porosity Map*.
 - **Connate Gas Saturation (Sg):** Populated via the *Initial Gas Saturation Map*.
- **Mathematical Constraints:** To prevent non-physical mathematical interpolation outside known control points, the "**Do not overshoot/undershoot points outside min/max values**" constraint was activated.
- **Grid Truncation (Nullification):** To isolate the reservoir volume from the inactive bounding geometry, grid blocks falling outside the defining structural contour boundary were deactivated.

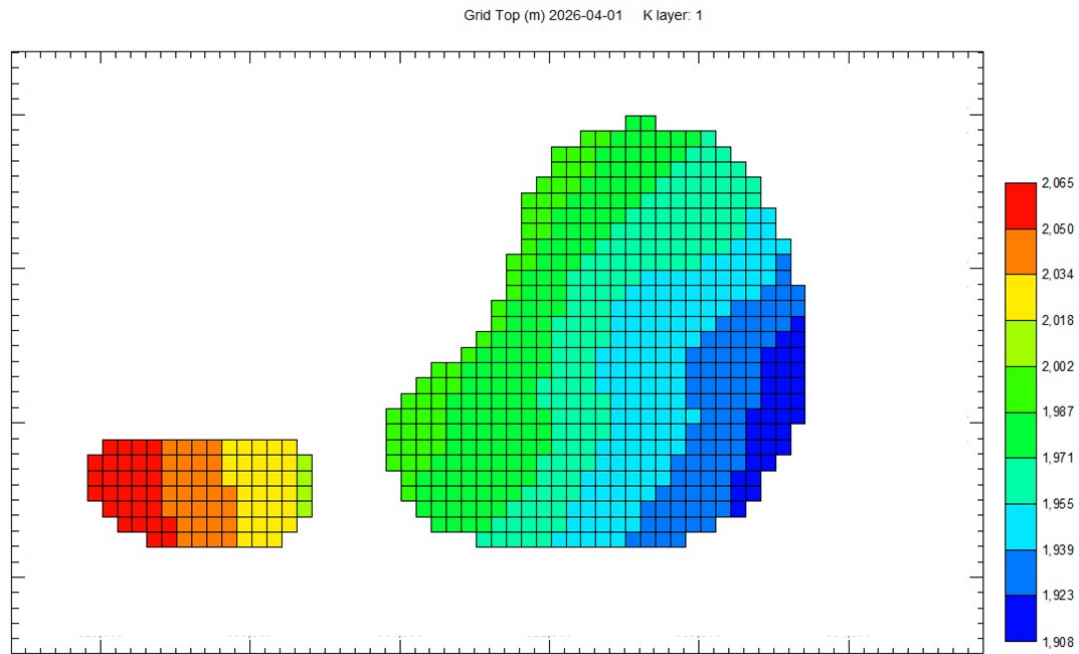


Figure 32: 2D Model of Grid Top

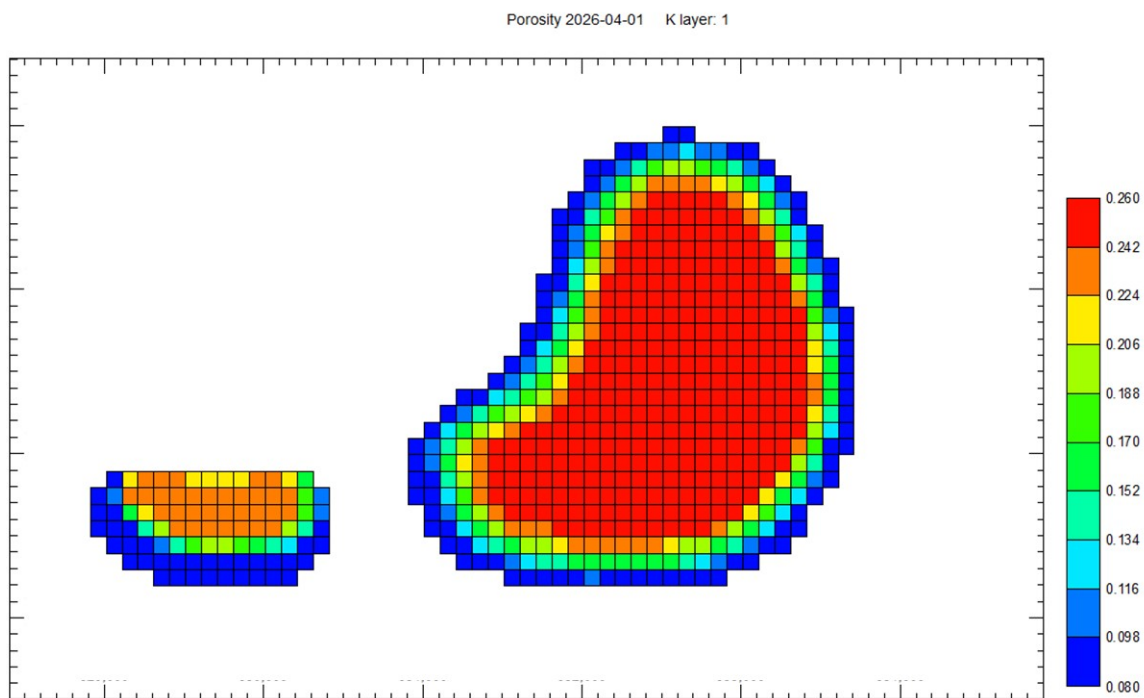


Figure 33: 2D Model of Porosity

Water Saturation 2026-04-01 K layer: 1

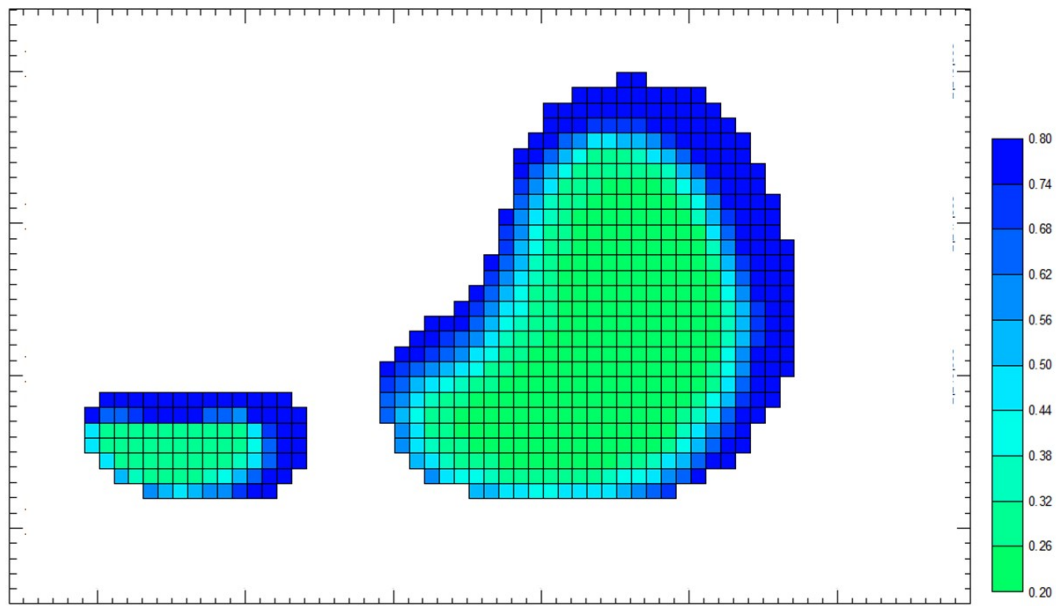


Figure 34: 2D Model of Saturation

5. Permeability Calculation

- Permeability was calculated using **Timur Equation**, where porosity and initial water saturation values are provided from log data :

$$k = a \cdot \frac{\phi^b}{S_{wi}^c} \quad k = 8581 \cdot \frac{\phi^{4.4}}{S_{wi}^2}$$

(Note: k is calculated in millidarcies [mD] and porosity and saturations are entered in fractions)

- Then, Phi – K graph was plotted and got the relation between porosity and permeability as
permeability = 9,64,358.780 x (porosity)^{5.187}

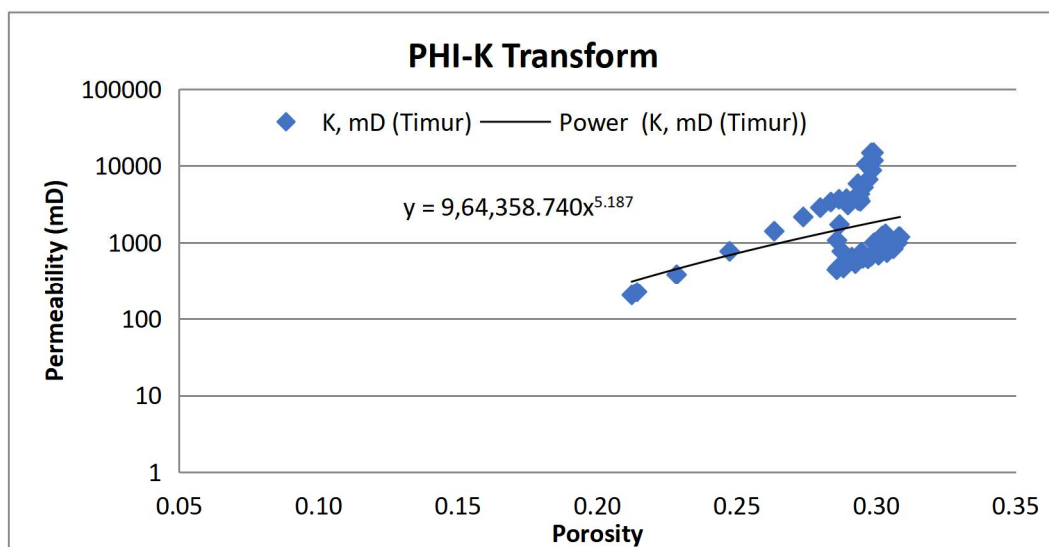


Figure 35: Phi -K Transform Plot

- **Formula Manager Configuration:** The user-defined relationship named PermX was created in the Formula Manager. Matrix porosity (ϕ) was defined as the independent driving variable. The calibrated Timur relationship was compiled mathematically into the manager and applied globally.

$$\text{PermX} = 964358.78 * (X0) ** 5.187$$

- **Directional Permeability Allocations:**
 - **Y-Direction Permeability (PermY):** Configured via the EQUALSI operator to mirror PermX (PermY = PermX).
 - **Z-Direction Permeability (PermZ):** Configured via the EQUALSI operator with a multiplier constraint of **0.1**. So that, the vertical anisotropy ratio (k_v/k_h) would be 0.1.

7. Formation Compressibility Configuration

- **Workflow:** Navigating through Reservoir → Rock Compressibility, the pressure dependence of formation porosity (CPOR) was specified as $5 \times 10^{-5} \text{ psi}^{-1}$ (or 1/psi).
- **Reference Pressure:** The reference pressure for calculating rock compressibility effects (PRPOR) was anchored to the initial reservoir condition of **340.529 KSC** (kg/cm²).
- **DCPOR** = 5×10^{-7}

8. Fluid Model Definition (PVT Correlations)

A representative black-oil fluid model was generated to establish gas and water phase behaviour across varying pressure regimes:

- **Workflow:** Navigating to Components → Component Properties → MODEL, the 'Launch Dialogue to create a Quick GAS/WATER Model Using Correlations' utility was executed.
- **Input Parameters:** The property matrix was populated using specific field data:
 - **Reservoir Temperature:** 103.5 °C
 - **Maximum Operating Pressure:** 350 KSC
 - **Gas Gravity:** 0.65 (relative to air)
 - **Reference Pressure for Water Properties:** 340.529 KSC
 - **Water Salinity:** 5000 ppm
- **PVT Curve Generation:** Based on these correlations, the simulator synthesized the curves including the Gas Formation Volume Factor (B_g) vs. Pressure (P) and Gas Viscosity (μ) vs. Pressure (P).

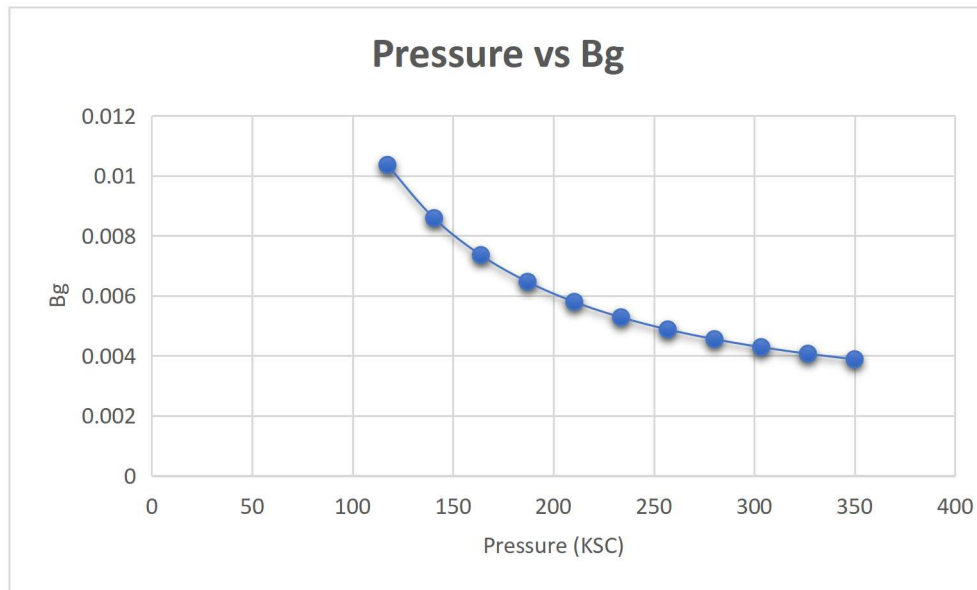


Figure 36: Pressure vs Bg Plot

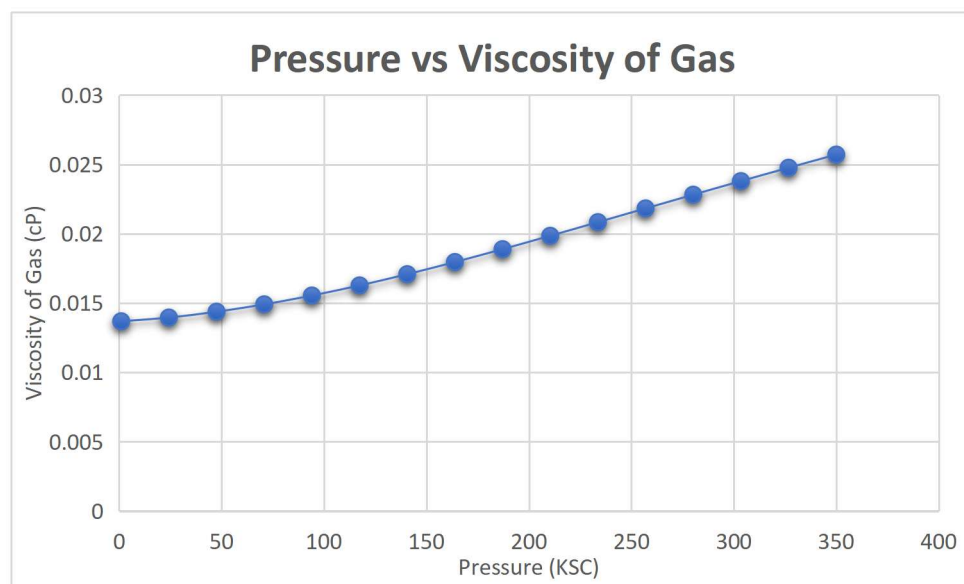


Figure 37: Gas Viscosity with Pressure

9. Rock-Fluid Interaction and Relative Permeability

Multiphase flow characteristics within the porous media were constrained by establishing relative permeability curves:

- **Workflow:** Under Rock Fluid → Rock Fluid Types → New Rock Type, a **Gas-Water system table** was selected to govern the displacement dynamics.

- **Parameters Input:** Relative permeability and saturation data were provided to generate the relative permeability curve.

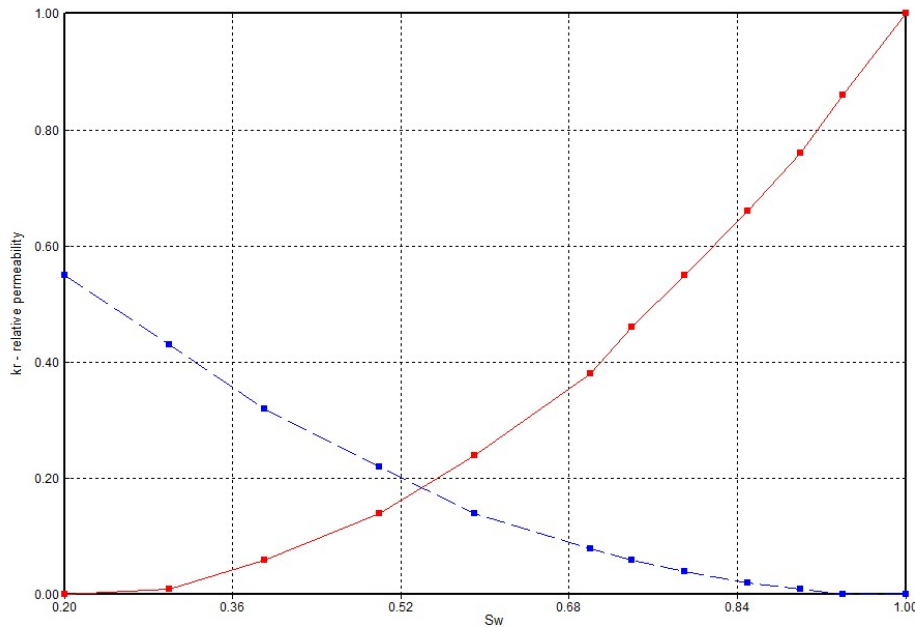


Figure 38: Relative Permeability Curve

10. Initialization of Reservoir Conditions

- **Equilibrium Method:** Under Initial Conditions → Initial Condition, the '*Perform Gravity-Capillary Equilibrium of a Reservoir Initially Containing Water and Gas*' calculation routine was initiated.
- **Input parameters:**
 - Reference pressure: **340.529 KSC**
 - Reference depth: **1992 m.**
 - Depth to Water-Gas Contact (**DWGC**): **2000 m.**

11. Model Validation, Initialization, and Volumetric Assessment

- **Dataset Validation:** The input deck was parsed via IMEX → Validate (Run to validate dataset only) to verify syntax correctness and check for missing arrays, grid compliance, or petrophysical out-of-bound errors.
- **Initialization Run:** Following a successful zero-error validation, a single time-step simulation initialization was executed (IMEX → Run to view initialization).
- **Volumetric Estimates (GIIP / WIIP):**

- **Total Gas Initial in Place (GIIP):** $0.31048 \times 10^{10} \text{ m}^3$ (approximately **3.1048 BCM** or Billion Cubic Meters).

12. Vertical Grid Refinement (Grid Splitting)

- **Workflow:** The grid editing toolkit was accessed via the Edit Grid interface.
- **Layer Discretization:** Navigating to Reservoir → Edit Grid → Split Grid Place, the primary vertical layer (K-direction) was selected and no of layers were set as 10.

13. Well Construction and Perforation Interval Configuration

- **Workflow:** Wells → New Well (specifying operational constraints, well type.
For the existing well framework, the perforation intervals were configured as follows:
- **Perforation Allocation:** Navigating to Wells → Well Completion (PERFS) → Perforation, target grid blocks were highlighted based on specific log depth intervals.
- **Reference Layer Assignment:** The shallowest active perforated layer was designated as the **Reference Layer** for bottomhole pressure calculation adjustments.

14. Dynamic Well Event Scheduling and Production Constraints

- **Workflow:** Wells → Events:
- One well was placed and made to flow with the actual well testing rates obtained from the exploratory well drilled in this reservoir.

Event Sequence	Phase	Date & Time	Operational Status	Constraints & Parameters
1	First Drawdown Period	02/04/2026 00:00:00 hrs	Open	Primary Operator: Surface Gas Rate (STG) Limit: Maximum Value: 1,44,441 m ³ /d
2	First Shut-in Period	02/04/2026 20:00:00 hrs	Shut-In	<i>All production constraints deleted.</i>

Event Sequence	Phase	Date & Time	Operational Status	Constraints & Parameters
3	Second Drawdown Period	03/04/2026 20:00:00 hrs	Open	Primary Operator: Surface Gas Rate (STG) Limit: Maximum Value: 95,628 m ³ /d
4	Second Shut-in Period	04/04/2026 05:00:00 hrs	Shut-In	<i>All production constraints deleted.</i>

15. Dynamic Execution and Simulation Post-Processing

- **Final Validation:** The comprehensive input deck was parsed using IMEX → Validate to rule out numerical conflicts or scheduling errors.
- **Simulation Run:** The full dynamic sequence was processed immediately via IMEX → Run Immediately.
- **Post-Processing (Results Viewer):** Upon successful completion of the simulation run, the output file was launched within the **CMG Results** module.

16. Model Calibration and History Matching (Drawdown Calibration)

- **Expected Drawdown Calculation:** Using the Flowing Tubing Head Pressure (FTHP) from the *Well Test Report* and a hydrostatic gradient of 0.1 KSC/10 m, the true field FBHP and expected drawdown (ΔP) were calculated as: **FBHP_{measured} = FTHP + 0.01 x Depth**

$$(\Delta P)_{\text{expected}} = P_R - \text{FBHP}_{\text{measured}}$$

We have FTHP = 4200 psi = 295.28 KSC and Depth till the mid of perforation = 1960.54 m

Calculated **FBHP_{measured} = 295.28 + 1960.54 x 0.01 = 314.9 KSC**

$$\text{Drawdown} = 340 - 314.9 = 25.1 \text{ KSC}$$

The simulation data was systematically calibrated to match this calculated drawdown baseline.

- **Skin Factor Adjustments:** A mechanical skin factor (s) of **31.5** was introduced in the Perforation window.
- **Permeability Tuning:** An additional factor of **0.11** was multiplied in the formula for permeability calculation.

- Model permeability restricted in the range of 1 to 96 mD.

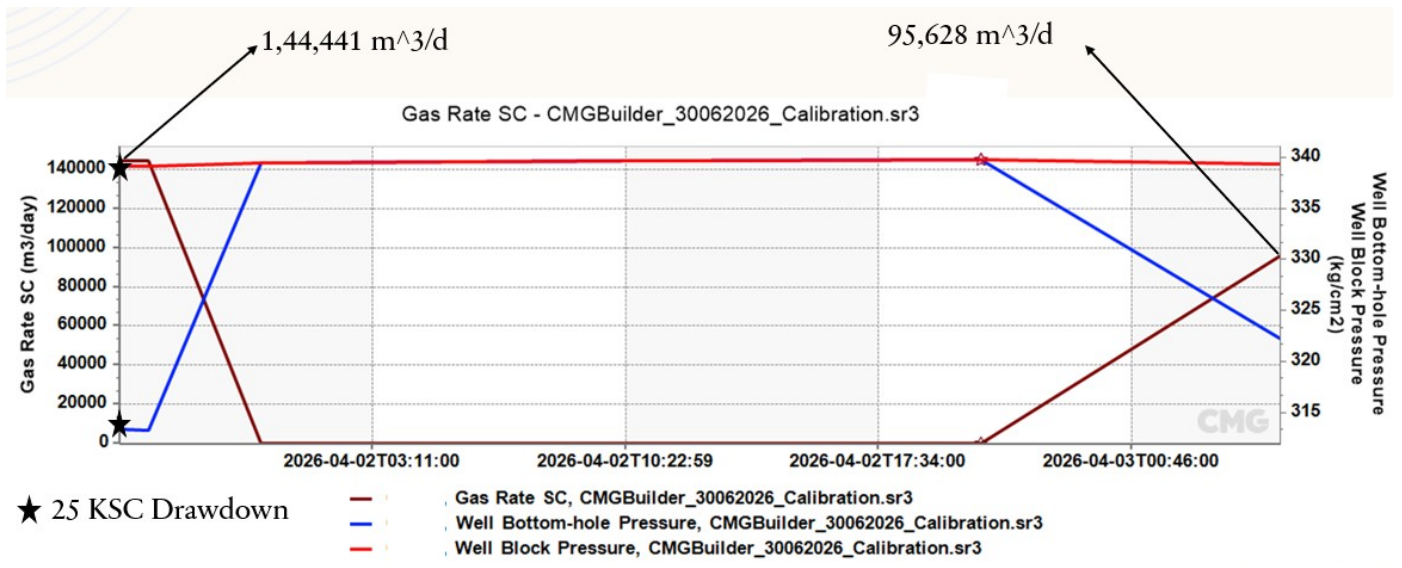


Figure 39: Simulation Results

17. Sector Modelling and Volumetric Compartmentalization

- **Workflow:** Navigating to Reservoir → Sector → Create/Edit Sector, new volumetric boundaries were established. Grid blocks were grouped laterally and extruded through the vertical column using the All layers (whole grid) constraint.
- **Sector Allocation:** The model was bifurcated into two specific regions: the **East Sector** and the **West Sector**.
- **Volumetric Contribution Analysis:** A dedicated simulation initialization run was executed to isolate the Gas Initial In Place (GIIP) for each domain. Consequently, the development strategy was optimized to allocate both planned production wells to the premium sector.

18. Development Well Placement and Completion Geometry

- **Placement Criteria:** Well locations were targeted in structurally favourable up-dip positions characterized by optimal matrix porosity and maximum net gas pay thickness.
- **Well Initialization:** Under Wells & Recurrent → Well → New, three producer well nodes were initialized with designated naming conventions and structural types.
- **Target Perforation Execution:** Using Wells → Well Completion (PERFS), the exact I, J grid block indices for each well location were specified via the Begin/Stop selection tool.

19. Production Inflow Events and Operational Constraints

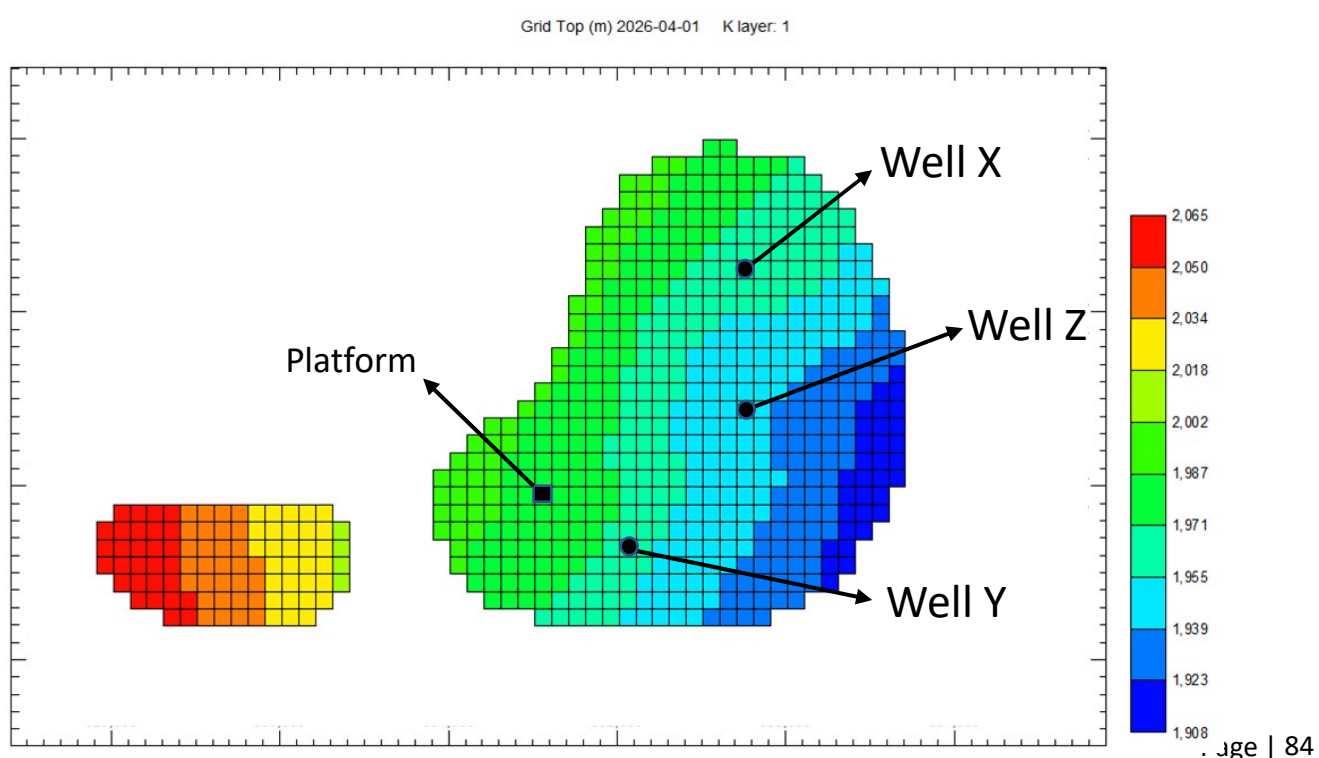
- **Workflow:** Configured under Well Events → [Well Name] → Type & ID as continuous producers with an active status of Open.
- **Operating Constraints:**
 - **Primary Control:** Maximum Surface Gas Rate (STG) capped at **1,50,000 m³/d**.
 - **Secondary Safety Control:** Minimum Flowing Bottomhole Pressure (FBHP) floor set at **120 KSC**.
- **Monitoring Constraints:** A lower-bound economic limit was established with a minimum monitor constraint of **35,000 m³/d (STG)**.

20. Production Forecasting and Simulation Timeline

- **Temporal Setup:** Navigating to Wells & Recurrent → Dates → Add Range of Dates, the forecasting window was formally scheduled from **01/04/2026** to a terminal date of **01/04/2040**.
- **Reporting Frequency:** The output step frequency was configured to **Monthly Reporting**.

21. Development Strategy

- We have done our analysis on a green field having GIIP of **3.1 BCM**, and decided to produce with three development wells. And we forecasted our model for the next **12 years** till we got economic production and getting the recovery factor of **49%**.



Production Profile of Well: X							
Year	Qg, MMSCMD	Qc, bcpd	Qw,bwpd	Cum Gas, BCM	Cum Cond, MMm3	SBHP, Ksc	FBHP, Ksc
2026-27	0.150	143	0.0	0.055	0.008	319	180
2027-28	0.150	143	0.0	0.110	0.017	295	153
2028-29	0.150	143	0.0	0.164	0.025	272	127
2029-30	0.141	134	0.0	0.216	0.033	250	122
2030-31	0.117	112	0.0	0.258	0.039	232	122
2031-32	0.097	92	0.0	0.294	0.045	216	122
2032-33	0.081	77	0.0	0.323	0.049	203	122
2033-34	0.068	65	0.0	0.348	0.053	193	122
2034-35	0.058	55	0.0	0.369	0.056	184	122
2035-36	0.050	48	0.0	0.388	0.059	177	122
2036-37	0.043	41	0.0	0.403	0.061	170	122
2037-38	0.037	36	0.0	0.417	0.063	165	122

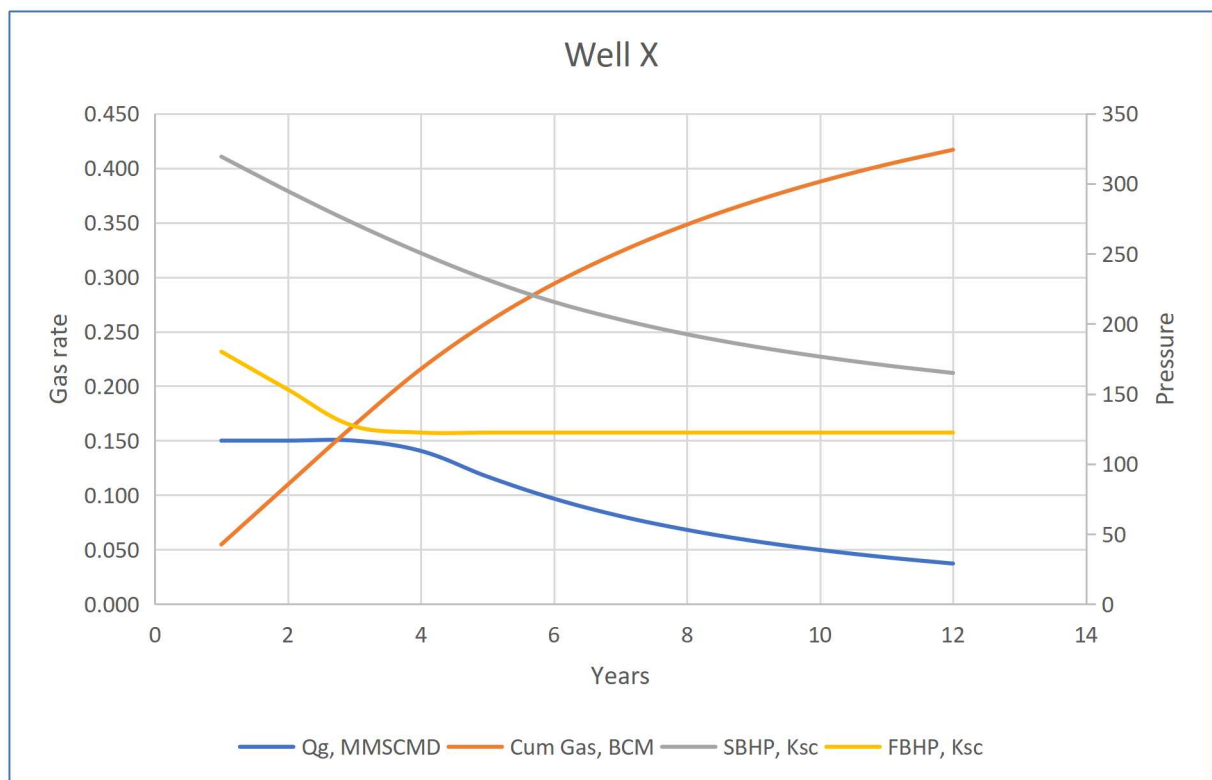


Figure 40: Production Profile of Well X

Production Profile of Well: Y							
Year	Q _g , MMSCMD	Q _c , bcpd	Q _w , bwpd	Cum Gas, BCM	Cum Cond, MMm3	SBHP, Ksc	FBHP, Ksc
2026-27	0.15	143	0	0.055	0.008	320	219
2027-28	0.15	143	0	0.110	0.017	296	192
2028-29	0.15	143	0	0.164	0.025	272	167
2029-30	0.15	143	0	0.219	0.033	251	143
2030-31	0.15	143	0	0.274	0.042	232	122
2031-32	0.13	127	0	0.323	0.049	216	122
2032-33	0.11	106	0	0.363	0.055	203	122
2033-34	0.09	90	0	0.397	0.060	193	122
2034-35	0.08	76	0	0.426	0.065	184	122
2035-36	0.07	65	0	0.452	0.069	177	122
2036-37	0.06	56	0	0.473	0.072	170	122
2037-38	0.05	49	0	0.492	0.075	165	122

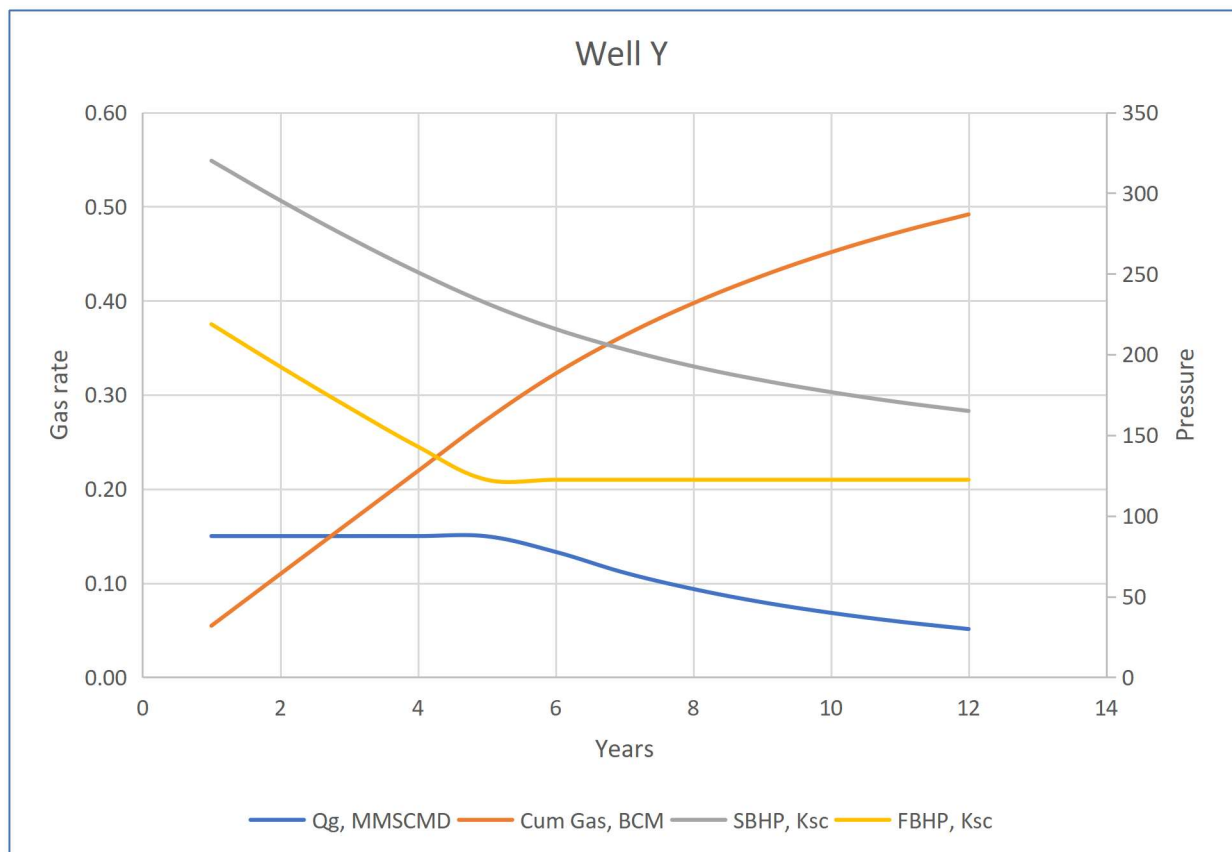


Figure 41: Production Profile of Well Y

Production Profile of Well: Z							
Year	Q _g , MMSCMD	Q _c , bcpd	Q _w , bwpd	Cum Gas, BCM	Cum Cond, MMm3	SBHP, Ksc	FBHP, Ksc
2026-27	0.15	143	0	0.055	0.008	319	218
2027-28	0.15	143	0	0.110	0.017	295	192
2028-29	0.15	143	0	0.164	0.025	272	166
2029-30	0.15	143	0	0.219	0.033	250	142
2030-31	0.15	143	0	0.274	0.042	231	122
2031-32	0.13	127	0	0.322	0.049	215	122
2032-33	0.11	106	0	0.363	0.055	203	122
2033-34	0.09	89	0	0.397	0.060	192	122
2034-35	0.08	76	0	0.426	0.065	184	122
2035-36	0.07	65	0	0.450	0.068	176	122
2036-37	0.06	56	0	0.472	0.072	170	122
2037-38	0.05	49	0	0.490	0.074	165	122

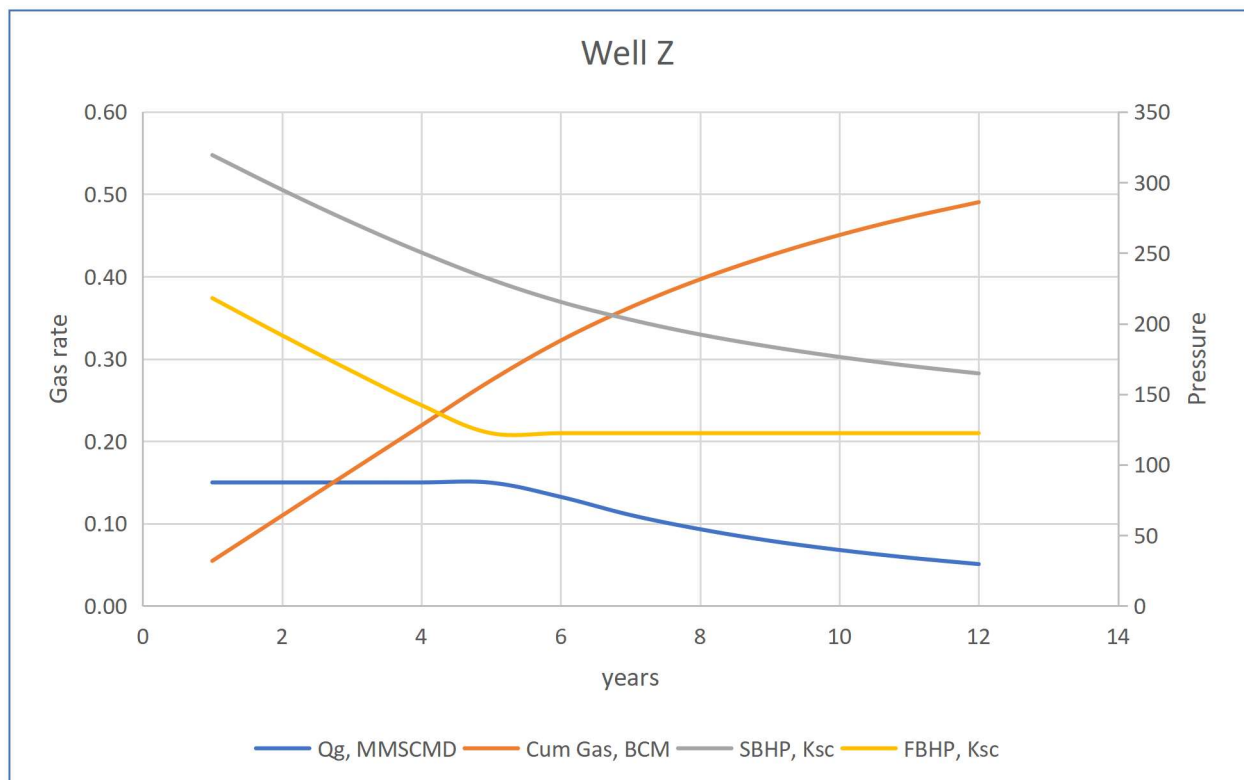


Figure 42: Production Profile of Well Z

Production Profile of Total Field					
Year	Qg, MMSCMD	Qc, bcpd	Qw,bwpd	Cum Gas, BCM	Cum Cond, MMm3
2026-27	0.45	430	0	0.164	0.025
2027-28	0.45	430	0	0.329	0.050
2028-29	0.45	430	0	0.493	0.075
2029-30	0.44	421	0	0.654	0.099
2030-31	0.42	398	0	0.806	0.122
2031-32	0.36	346	0	0.939	0.143
2032-33	0.30	289	0	1.049	0.159
2033-34	0.26	244	0	1.142	0.173
2034-35	0.22	207	0	1.221	0.186
2035-36	0.19	178	0	1.289	0.196
2036-37	0.16	153	0	1.348	0.205
2037-38	0.14	133	0	1.399	0.212

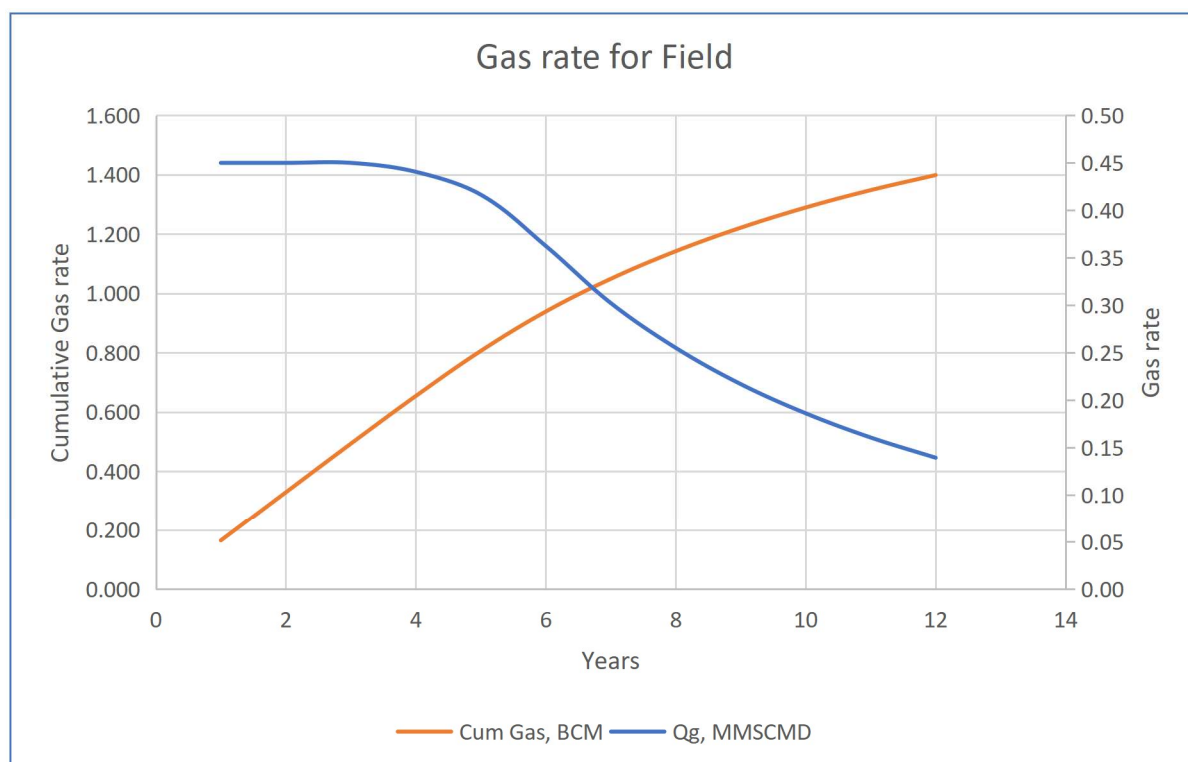


Figure 43: Production Profile of Total field